

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AUTONOMOUS DEGREE SEMESTER EXAMINATION

Subject: Design and Analysis of Algorithms	Sub Code: CS-501	Max Marks: 35
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions.
- 2. Assume suitable data if necessary.

PART A (SHORT ANSWER)

SHORT ANSWERS

1. Define Introduction: Algorithm and state its significance in Design and Analysis of Algorithms. **[2 Marks]**
2. Define Disjoint Sets: Disjoint set operations and state its significance in Design and Analysis of Algorithms. **[2 Marks]**
3. Define Dynamic Programming: General method and state its significance in Design and Analysis of Algorithms. **[2 Marks]**
4. Define Greedy method: General method and state its significance in Design and Analysis of Algorithms. **[2 Marks]**
5. Define Branch and Bound: General method and state its significance in Design and Analysis of Algorithms. **[2 Marks]**

PART B (DESCRIPTIVE THEORY & PROBLEMS)

UNIT-WISE ANALYTICAL QUESTIONS (WITH INTERNAL CHOICE)

6. (a) Describe the architectural or structural flow involved in Introduction: Algorithm. **[5 Marks]**
[OR]
(b) What is the primary function or objective of Introduction: Algorithm? **[5 Marks]**
7. (a) Compare and contrast the advantages and limitations of Disjoint Sets: Disjoint set operations. **[5 Marks]**
[OR]
(b) What is the primary function or objective of Disjoint Sets: Disjoint set operations? **[5 Marks]**
8. (a) Explain the fundamental working principle of Dynamic Programming: General method with suitable illustrations. **[5 Marks]**
[OR]
(b) What is the primary function or objective of Dynamic Programming: General method? **[5 Marks]**
9. (a) Illustrate the step-by-step mechanism of Greedy method: General method using a concrete example. **[5 Marks]**
[OR]
(b) What is the primary function or objective of Greedy method: General method? **[5 Marks]**

10. (a) Discuss how Branch and Bound: General method handles edge cases, boundary conditions, or exceptions. **[5 Marks]**

[OR]

(b) What is the primary function or objective of Branch and Bound: General method? **[5 Marks]**